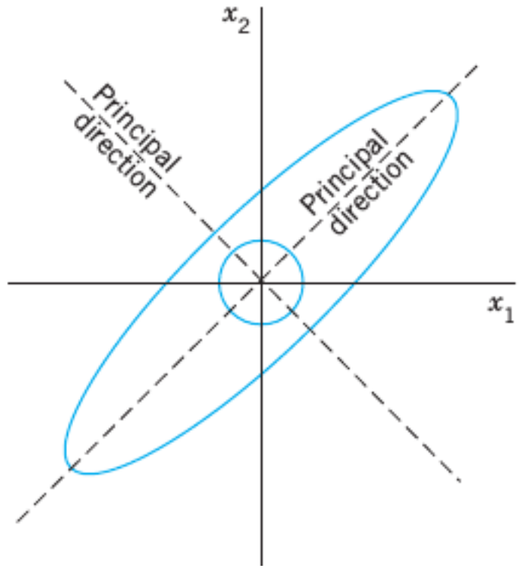
8.2 Some Applications of Eigenvalue Problems

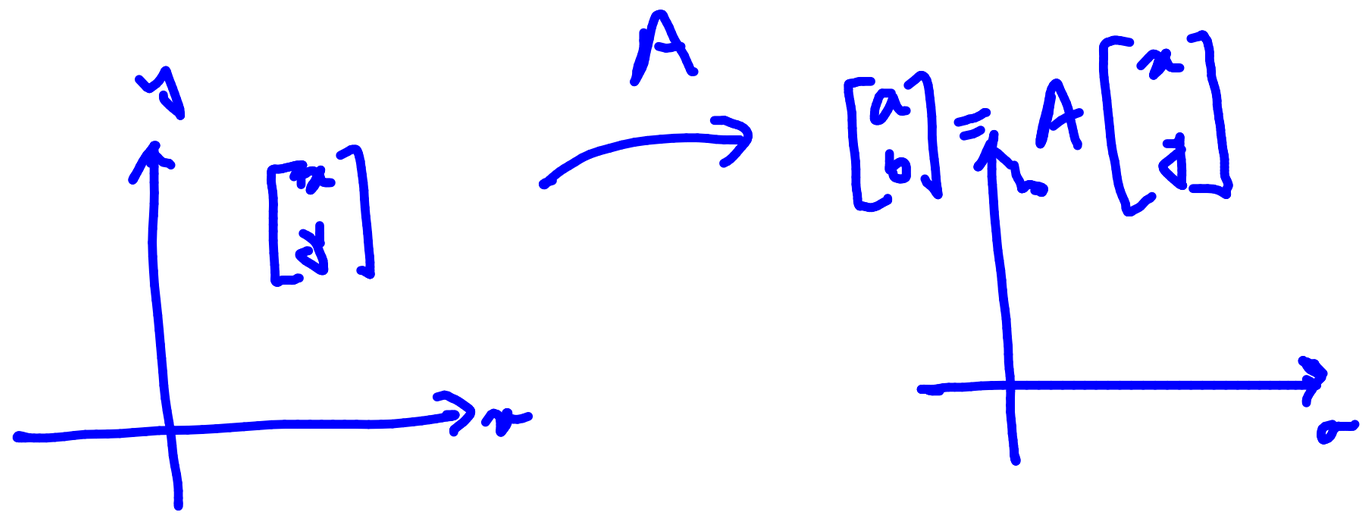
Stretching of an Elastic Membrane



Given $\mathbf{A}$ in a deformation $\mathbf{y} = \mathbf{A}\mathbf{x}$, find the principal directions and corresponding factors of extension or contraction. Show the details.

1. $\begin{bmatrix} 3.0 & 1.5 \\ 1.5 & 3.0 \end{bmatrix}$ **2.** $\begin{bmatrix} 2.0 & 0.4 \\ 0.4 & 2.0 \end{bmatrix}$

A : 2x2 matrix



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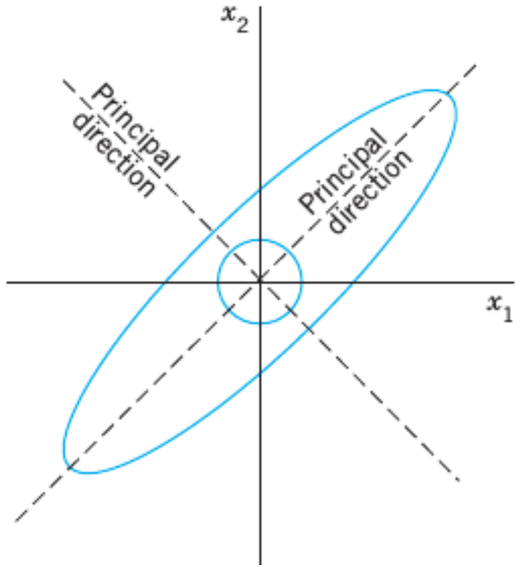

eigenvalues & eigenvectors

$\vec{v}$, λ

$A\vec{v} = \lambda\vec{v}$

Given **A** in a deformation $\mathbf{y} = \mathbf{A}\mathbf{x}$, find the principal directions and corresponding factors of extension or contraction. Show the details.

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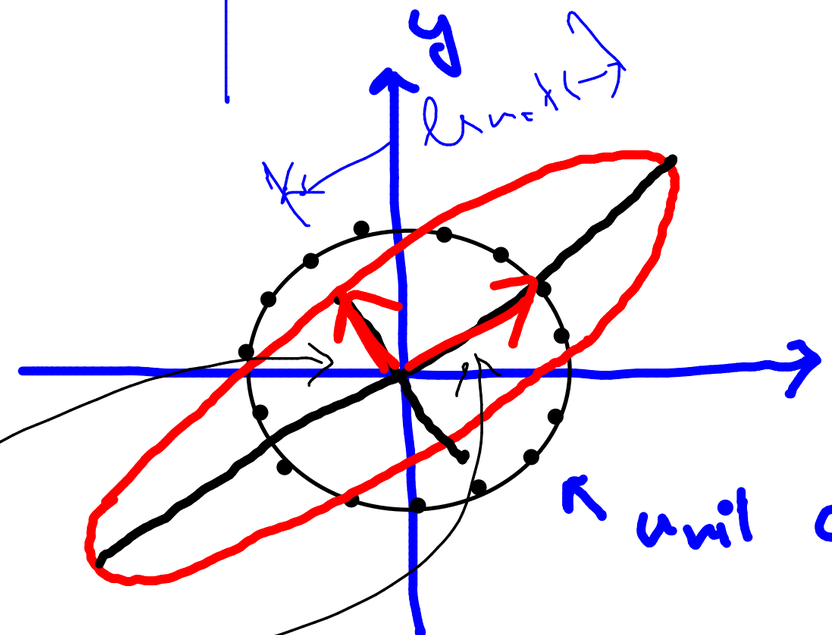
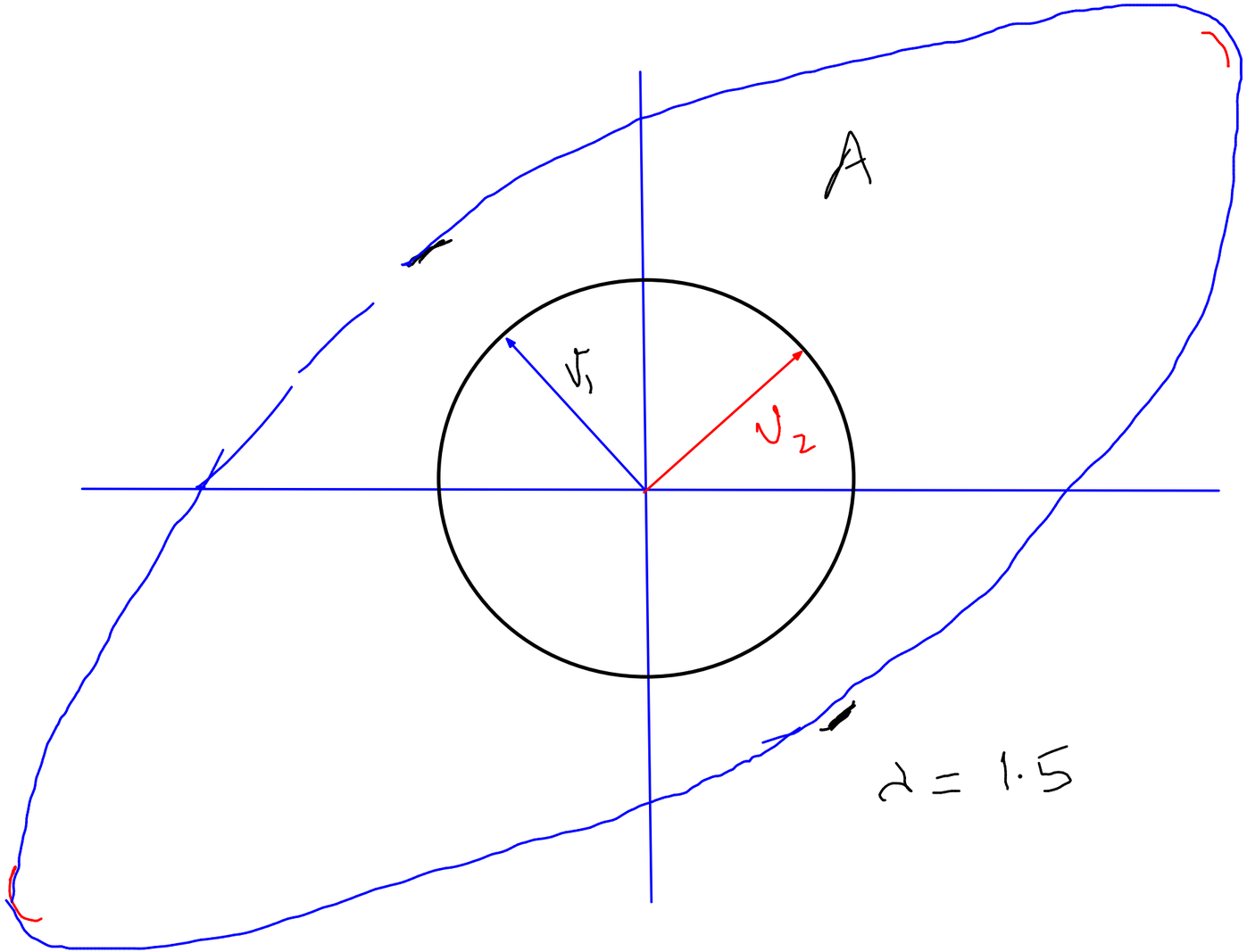


Principle directions

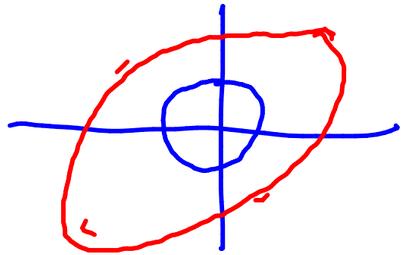
circle got transformed into an ellipse under the action of A




```
A =

    3.0000    1.5000
    1.5000    3.0000

>> [v, d] = eig(A)

v =

   -0.7071    0.7071
    0.7071    0.7071


d =

    1.5000         0
         0    4.5000
```

Given **A** in a deformation $\mathbf{y} = \mathbf{A}\mathbf{x}$, find the principal directions and corresponding factors of extension or contraction. Show the details.

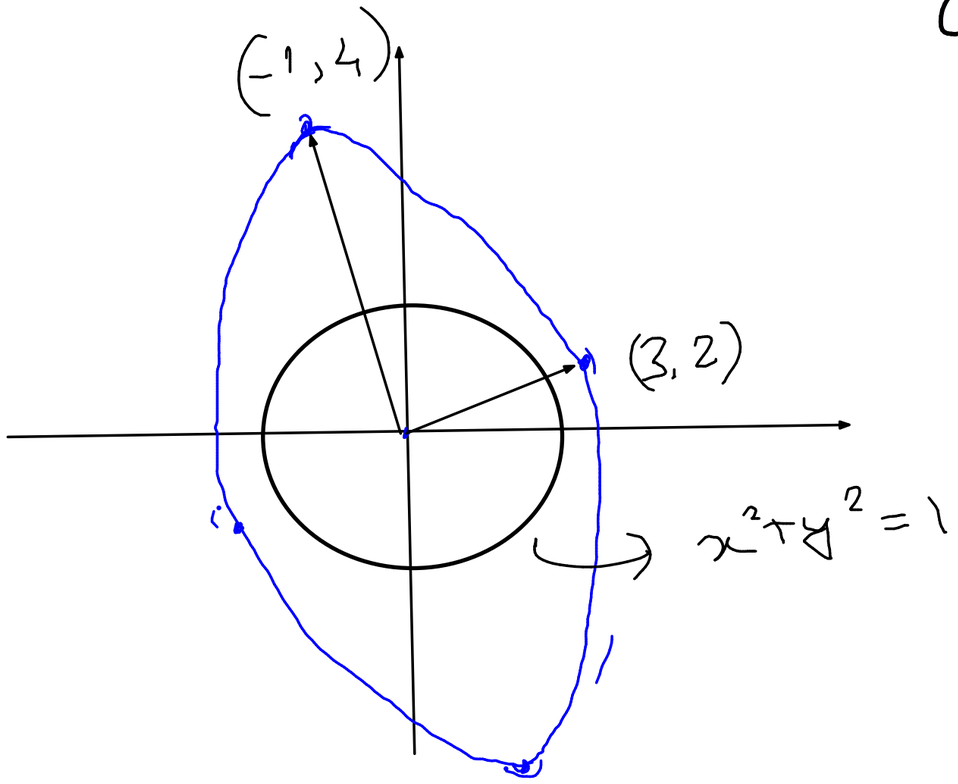
$$\begin{bmatrix} 2.0 & 0.4 \\ 0.4 & 2.0 \end{bmatrix}$$



$\lambda = 1.6$ & $v = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

$\lambda = 2.4$, $v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

```
A =

   2.0000    0.4000
   0.4000    2.0000

>> [v d] = eig(A)

v =

  -0.7071    0.7071
   0.7071    0.7071


d =

   1.6000         0
        0    2.4000
```



Question

find eigenvalue & corresponding eigenvector

$$\vec{v}_1 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \quad \alpha = \sqrt{13}$$

$$\vec{v}_2 = \begin{bmatrix} -1 \\ 4 \end{bmatrix}, \quad \alpha = \sqrt{17}$$

Eigenvalue Problems Arising from Population Models. Leslie Model

The Leslie model describes age-specified population growth, as follows. Let the oldest age attained by the females in some animal population be 9 years. Divide the population into three age classes of 3 years each. Let the "*Leslie matrix*" be

a)

$$\begin{bmatrix} 0 & 2.3 & 0.4 \\ 0.6 & 0 & 0 \\ 0 & 0.3 & 0 \end{bmatrix} \begin{bmatrix} 400 \\ 400 \\ 400 \end{bmatrix} = \begin{bmatrix} \text{population} \\ \text{kids} \\ 400(0.6) \\ 400(0.3) \end{bmatrix}$$

(5) $$\mathbf{L} = [l_{jk}] = \begin{bmatrix} 0 & 2.3 & 0.4 \\ 0.6 & 0 & 0 \\ 0 & 0.3 & 0 \end{bmatrix}$$

where l_{1k} is the average number of daughters born to a single female during the time she is in age class k, and $l_{j,j-1} (j = 2, 3)$ is the fraction of females in age class $j - 1$ that will survive and pass into class j. (a) What is the number of females in each class after 3, 6, 9 years if each class initially consists of 400 females? (b) For what initial distribution will the number of females in each class change by the same proportion? What is this rate of change?

b)

$$\begin{bmatrix} 0 & 2.3 & 0.4 \\ 0.6 & 0 & 0 \\ 0 & 0.3 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \lambda \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$\begin{bmatrix} x \\ y \\ z \end{bmatrix}$: is an eigenvector

λ : is an eigenvalue

```
v =

   -0.8889
   -0.4444
   -0.1111


d =

    1.2000
         0   -
         0
```

$$v = \begin{bmatrix} .8889 \\ .4444 \\ .1111 \end{bmatrix} \xrightarrow[\text{integer}]{\text{scale to}}$$

population after 3 years

$$\begin{bmatrix} 0 & 2.3 & 0.4 \\ 0.6 & 0 & 0 \\ 0 & 0.1 & 0 \end{bmatrix} \begin{bmatrix} 400 \\ 400 \\ 400 \end{bmatrix}$$

population after 6 years

$$\begin{bmatrix} 0 & 2.3 & 0.4 \\ 0.6 & 0 & 0 \\ 0 & 0.1 & 0 \end{bmatrix}^2 \begin{bmatrix} 400 \\ 400 \\ 400 \end{bmatrix}$$

$$L = \begin{bmatrix} 0 & 2.3 & 0.4 \\ 0.6 & 0 & 0 \\ 0 & 0.3 & 0 \end{bmatrix} \qquad P = \begin{array}{c} \text{population in} \\ \begin{bmatrix} 100 \\ 400 \\ 200 \end{bmatrix} \end{array} \begin{array}{l} \leftarrow \text{age group 1} \\ \leftarrow \text{age group 2} \\ \leftarrow \text{age group 3} \end{array}$$

Q: find population after three years

$$\begin{bmatrix} 0 & 2.3 & 0.4 \\ 0.6 & 0 & 0 \\ 0 & 0.3 & 0 \end{bmatrix} \begin{bmatrix} 100 \\ 400 \\ 200 \end{bmatrix}$$

13. Leontief input–output model. Suppose that three industries are interrelated so that their outputs are used as inputs by themselves, according to the 3×3 **consumption matrix**

$$\mathbf{A} = [a_{jk}] = \begin{bmatrix} 0.1 & 0.5 & 0 \\ 0.8 & 0 & 0.4 \\ 0.1 & 0.5 & 0.6 \end{bmatrix}$$

where a_{jk} is the fraction of the output of industry k consumed (purchased) by industry j. Let p_j be the price charged by industry j for its total output. A problem is to find prices so that for each industry, total expenditures equal total income. Show that this leads to $\mathbf{Ap} = \mathbf{p}$, where $\mathbf{p} = [p_1 \quad p_2 \quad p_3]^\mathsf{T}$, and find a solution $\mathbf{p}$ with nonnegative p_1, p_2, p_3.

your job.